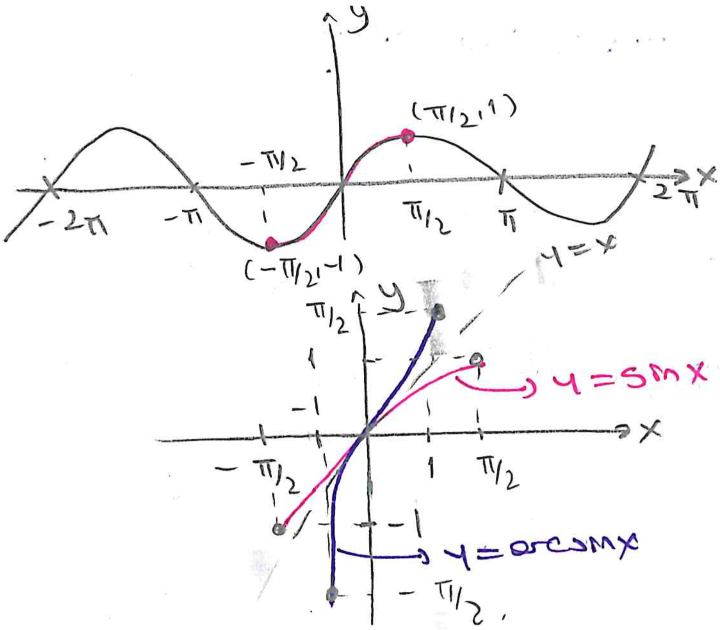
MATH 123 LECTURE NOTES

WEEK III

THE INVERSE TRIGONOMETRIC FUNCTIONS

THE INVERSE SINE (OR ARCSINE) FUNCTION



$y=\sin x$ is 1-1 for $x \in [-\pi/2, \pi/2]$

This means that we obtain its inverse in this interval.

$\sin x : [-\pi/2, \pi/2] \longrightarrow [-1,1]$

$\arcsin x : [-1,1] \longrightarrow [-\pi/2, \pi/2]$

$$\boxed{\begin{array}{lll} y=\arcsin x \iff x = \sin y & y\in[-\pi/2, \pi/2] & x \in [-1,1] \\ \sin(\arcsin x) = x \quad x\in[-1,1], & \arcsin(\sin x) = x & x \in [-\pi/2, \pi/2] \end{array}}$$

Ex 7 Find
ⓐ $\sin(\arcsin 0.7)$ ⓑ $\arcsin(\sin 0.3)$ ⓒ $\arcsin(\sin \frac{4\pi}{5})$
ⓓ $\cos(\arcsin 0.6)$ ⓔ $\tan(\arcsin x)$

ⓐ $\sin(\arcsin(0.7)) = 0.7$ ⓑ $\arcsin(\sin 0.3) = 0.3$

ⓒ $\arcsin(\sin \frac{4\pi}{5})$ $\quad \frac{4\pi}{5} = \frac{8\pi}{10} > \frac{\pi}{2}$ Therefore $\frac{4\pi}{5} \notin [-\pi/2, \pi/2]$
which means that we must convert $\frac{4\pi}{5}$ into an angle in $[-\pi/2, \pi/2]$

$\sin(\frac{4\pi}{5}) = \sin(\pi - \pi/5) = \sin \pi/5$ $\qquad \pi/5 \in [-\pi/2, \pi/2]$

$\arcsin(\sin(\frac{4\pi}{5})) = \arcsin(\sin \pi/5) = \pi/5$

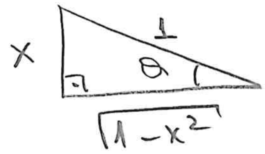
ⓓ $\cos(\underbrace{\arcsin 0.6}_{\theta}) = ?$ $\qquad \arcsin 0.6 = \theta$
$\qquad \sin\theta = 0.6$
$\cos(\arcsin 0.6) = 0.8$ $\qquad 0.8 = \sqrt{1-(0.6)^2}$

(1)

(e) $\tan(\arcsin x) = ?$

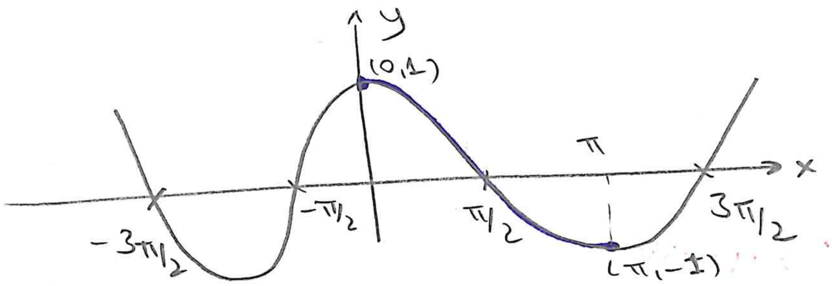
$\arcsin x = \theta$ $\quad$ $\sin\theta = x$

$\arcsin x = \theta$ $\quad$ $\tan\theta = ?$



$\tan\theta = \dfrac{x}{\sqrt{1-x^2}}$.

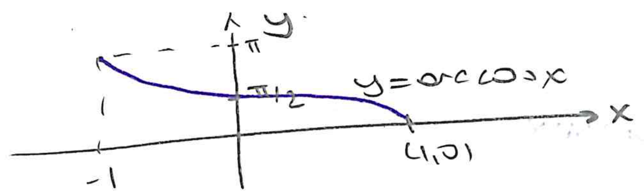
THE INVERSE COSINE FUNCTIONS ($\arccos x$)



- $y = \cos x$ is 1−1 for $x \in [0,\pi]$

$\cos x : [0,\pi] \mapsto [-1,1]$

- $\arccos x : [-1,1] \mapsto [0,\pi]$

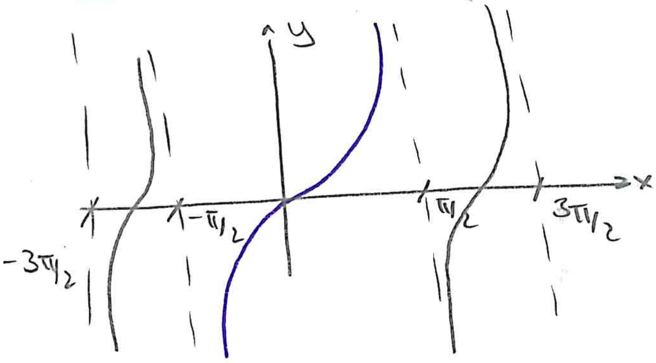


$$\boxed{\begin{array}{l} y = \arccos x \iff x = \cos y \qquad y \in [0,\pi],\ x \in [-1,1] \\ \cos(\arccos x) = x \quad x \in [-1,1],\ \arccos(\cos x) = x \quad x \in [0,\pi] \end{array}}$$

$y = \arccos x \iff x = \cos y = \sin(\pi/2 - y) \iff \arcsin x = \frac{\pi}{2} - y = \frac{\pi}{2} - \arccos x$

$$\boxed{\arccos x = \frac{\pi}{2} - \arcsin x \quad \text{for } -1 \le x \le 1}$$

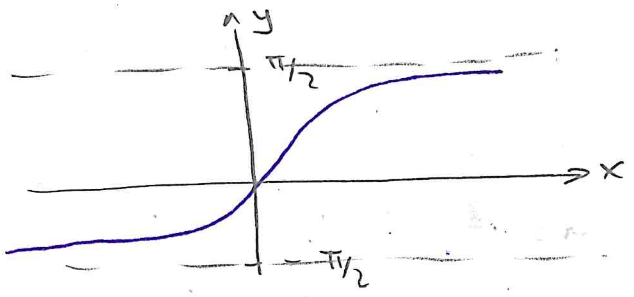
THE INVERSE TANGENT (arctan x) FUNCTION



$y = \tan x \quad x \in (-\pi/2, \pi/2) \quad y \in (-\infty, \infty)$

1−1 in this time interval

$\tan x : (-\pi/2, \frac{\pi}{2}) \to (-\infty, \infty)$

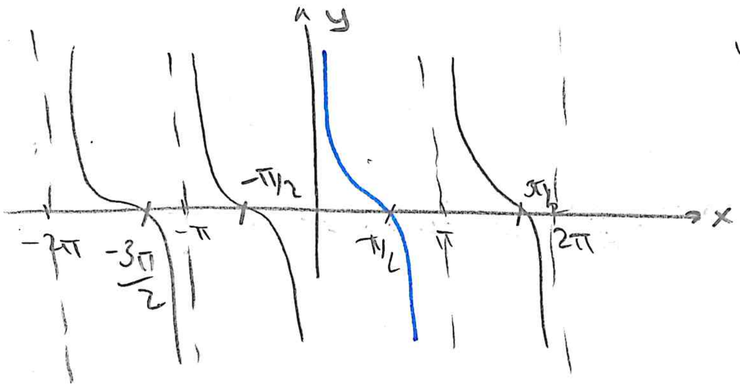


$y = \arctan x \quad \arctan x : (-\infty, \infty) \longmapsto (-\pi/2, \pi/2)$

$y = \arctan x \iff x = \tan y \quad y \in (-\pi/2, \pi/2) \quad x \in (-\infty, \infty)$

$\tan(\arctan x) = x \quad x \in (-\infty, \infty), \quad \arctan(\tan x) = x \quad x \in (-\pi/2, \pi/2)$

THE INVERSE COTANJENT FUNCTION (arc cotx)

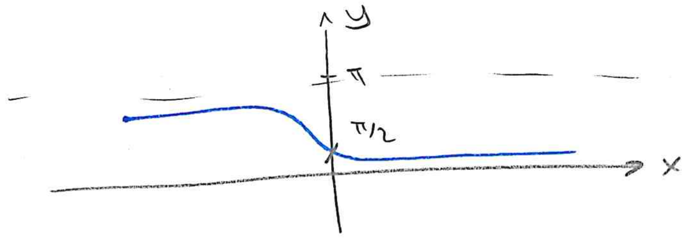


$y = \cot x \quad x \in (0, \pi) \quad y \in (-\infty, \infty)$

$\cot x : (0, \pi) \longmapsto (-\infty, \infty)$

$y = \operatorname{arccot} x$

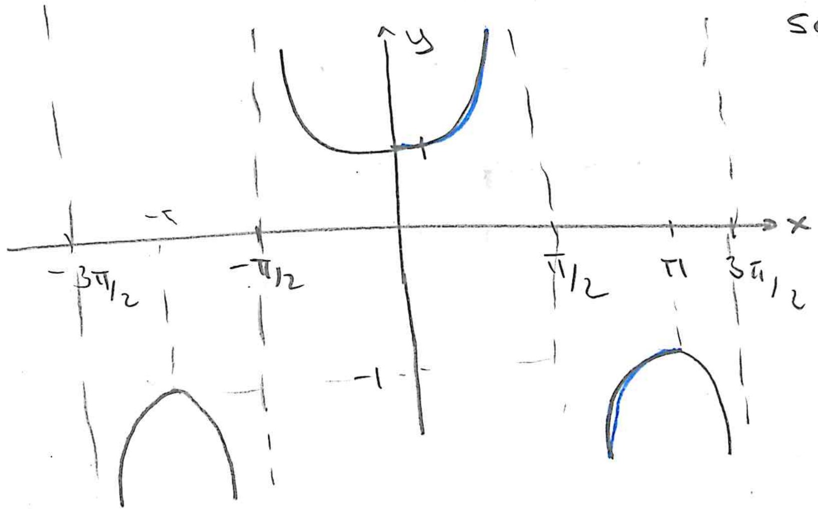
$\operatorname{arccot} x : (-\infty, \infty) \to (0, \pi)$



$y = \operatorname{arccot} x \iff x = \cot y \quad x \in (-\infty, \infty) \quad y \in (0, \pi)$

$\operatorname{arccot}(\cot x) = x \quad x \in (0, \pi), \quad \cot(\operatorname{arccot} x) = x \quad x \in (-\infty, \infty)$

THE INVERSE SECANT FUNCTION (arcsecx)

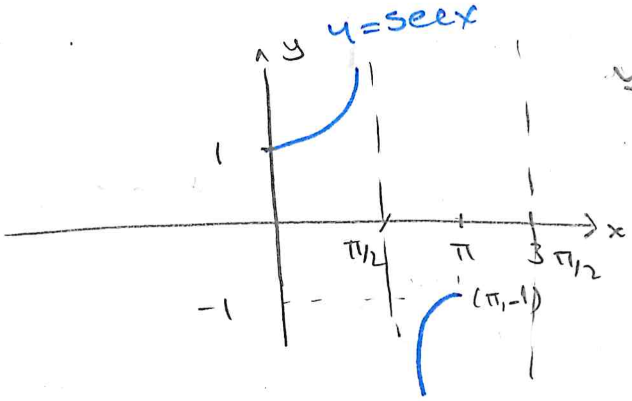


$\sec x = \frac{1}{\cos x}$

$\sec x$ 1-1 in the interval

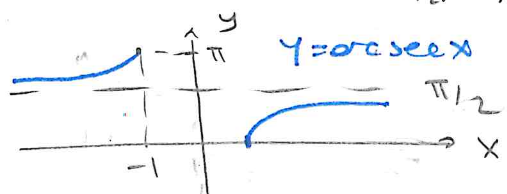
$[0, \pi/2) \cup (\pi/2, \pi]$

$\sec x : [0,\pi/2) \cup (\pi/2, \pi] \to (-\infty, -1] \cup [1, \infty)$



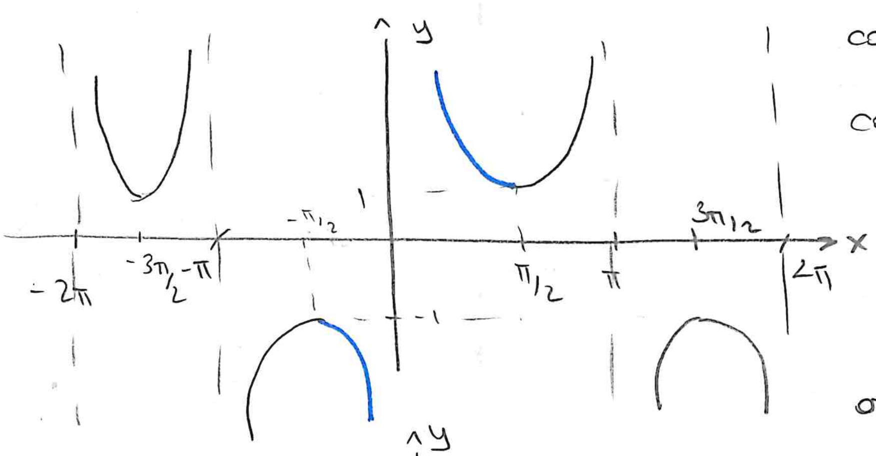
$y = \sec x$

$\operatorname{arcsec} x : (-\infty, -1] \cup [1, \infty) \to [0, \pi/2) \cup (\pi/2, \pi]$



$$\boxed{\begin{aligned} &y = \operatorname{arcsec} x \iff x = \sec y \quad x \in (-\infty, -1] \cup [1, \infty) \\ &\sec(\operatorname{arcsec} x) = x \quad x \in (-\infty, -1] \cup [1, \infty) \\ &\operatorname{arcsec}(\sec x) = x \quad x \in [0, \pi/2) \cup (\pi/2, \pi] \end{aligned}}$$

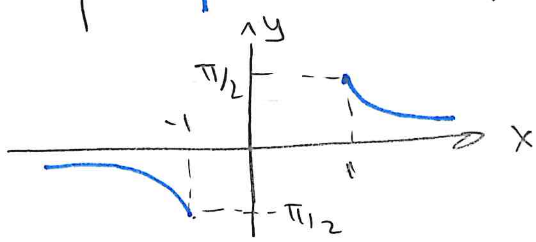
THE INVERSE COSECANT FUNCTION (arccosecx)



$\operatorname{cosec} x = \frac{1}{\sin x}$

$\operatorname{cosec} x : [-\pi/2, 0) \cup (0, \pi/2] \to (-\infty, -1] \cup [1, \infty)$

$\operatorname{arccos} x : (-\infty, -1] \cup [1, \infty) \to [-\pi/2, 0) \cup (0, \pi/2]$



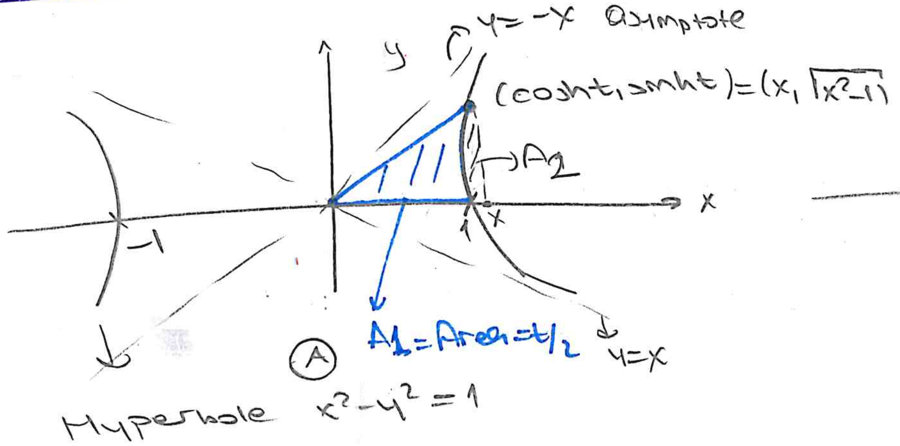
$$y = \operatorname{arccosec} x \iff x = \operatorname{arccosec} y \qquad y \in [-\pi/2, 0) \cup (0, \pi/2]$$
$$x \in (-\infty, -1] \cup [1, \infty)$$

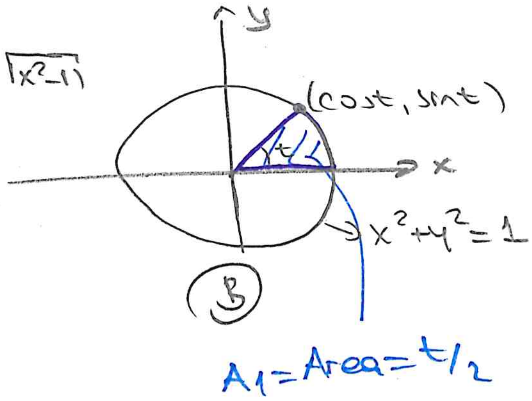
$$\operatorname{cosec}(\operatorname{arccosec} x) = x \qquad x \in (-\infty, -1] \cup [1, \infty)$$

$$\operatorname{arccosec}(\operatorname{cosec} x) = x \qquad x \in [-\pi/2, 0) \cup (0, \pi/2]$$

HYPERBOLIC FUNCTIONS

Trigonometric Functions versus Hyperbolic functions





Area of dot $\pi r^2 = \pi$

A_1 area $\pi \cdot \frac{t}{2\pi} = t/2$

How to obtain A_1 in graph (A)

$A_1 + A_2 =$ Area of the triangle with vertices $= (0,0), (x,0), (x, \sqrt{x^2-1})$

$$= \frac{x \cdot \sqrt{x^2-1}}{2}$$

The area of $A_2 = \int_1^x (\sqrt{x^2-1} - 0)\,dx = \frac{1}{2}x\sqrt{x^2-1} - \frac{1}{2}\ln(x + \sqrt{x^2-1})$

$$A_1 = \frac{1}{2}x \cdot \sqrt{x^2-1} - \int_1^x \sqrt{x^2-1}\,dx = \frac{1}{2}\ln(x + \sqrt{x^2-1})$$

$$e^{2A_1} = x + \underbrace{\sqrt{x^2-1}}_{y} \qquad e^{2A_1} = x + y \qquad e^{2A_1}(x-y) = (x+y)(x-y)$$

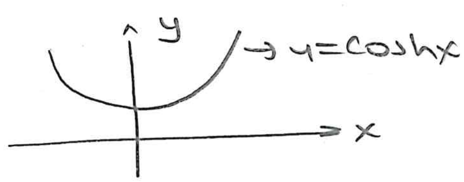
$$e^{2A_1}(x-y) = x^2 - y^2 = 1 \qquad x - y = e^{-2A_1}$$

$$x = \underbrace{\frac{e^{-2A_1} + e^{2A_1}}{2}}_{\cosh 2A_1} \qquad y = \underbrace{\frac{e^{-2A_1} - e^{2A_1}}{2}}_{\sinh 2A_1}$$

coshx

$$\cosh x = \frac{e^{x}+e^{-x}}{2}$$

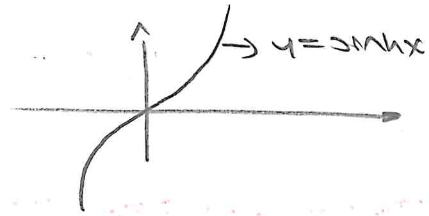
$\cosh x : (-\infty,\infty) \longmapsto [1,\infty)$



sinhx

$$\sinh x = \frac{e^{x}-e^{-x}}{2}$$

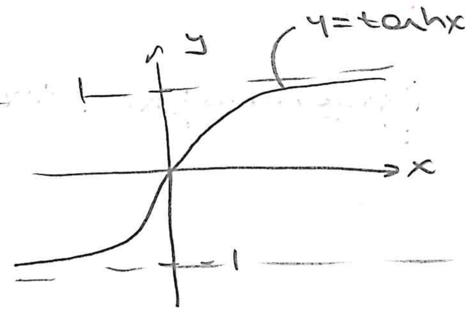
$\sinh x : (-\infty,\infty) \longmapsto (-\infty,\infty)$



tanhx

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$$

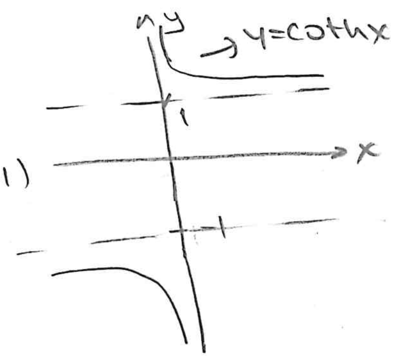
$\tanh x : (-\infty,\infty) \longmapsto (-1,1)$



cothx

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^{x}+e^{-x}}{e^{x}-e^{-x}}$$

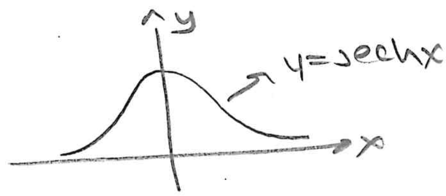
$\coth x : (-\infty,\infty)\setminus\{0\} \longmapsto (1,\infty) \cup (-\infty,-1)$



sechx

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^{x}+e^{-x}}$$

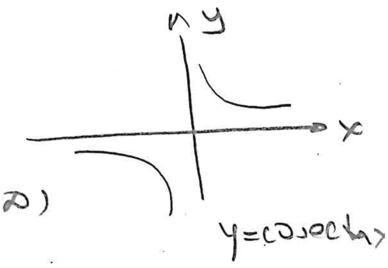
$\operatorname{sech} x : (-\infty,\infty) \longmapsto (0,1]$



cosechx

$$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^{x}-e^{-x}}$$

$\operatorname{cosech} x \; (-\infty,0) \cup (0,\infty) \longmapsto (-\infty,0) \cup (0,\infty)$



Properties of Hyperbolic Functions

① $\sinh(-x) = -\sinh x$ ② $\cosh(-x) = \cosh x$

③ $\tanh(-x) = -\tanh x$ ④ $\coth(-x) = -\coth x$

⑤ $\cosh^2 x - \sinh^2 x = 1$ ⑥ $1 - \tanh^2 x = \operatorname{sech}^2 x$

⑦ $\coth^2 x - 1 = \operatorname{cosech}^2 x$ ⑧ $\sinh^2 x = \frac{1}{2}(-1 + \cosh 2x)$

⑨ $\cosh^2 x = \frac{1}{2}(1 + \cosh 2x)$

⑩ $\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$

⑪ $\sinh(x-y) = \sinh x \cosh y - \cosh x \sinh y$

⑫ $\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$

⑬ $\cosh(x-y) = \cosh x \cosh y - \sinh x \sinh y$

⑭ $\sinh 2x = 2 \sinh x \cosh x$

⑮ $\cosh 2x = \cosh^2 x + \sinh^2 x$

Ex Show that $\sinh(-x) = -\sinh x$

$$\sinh(-x) = \frac{e^{-x} - e^{-(-x)}}{2} = \frac{e^{-x} - e^{x}}{2} = -\underbrace{\frac{e^{x} - e^{-x}}{2}} = -\sinh x$$

Ex $\sinh x = 3/4$ obtain $x = ?$

$$\sinh x = \frac{e^x - e^{-x}}{2} = 3/4 \Rightarrow 2e^x - 2e^{-x} = 3$$

Let $e^x = a$ $\quad 2a - \frac{2}{a} = 3 \Rightarrow 2a^2 - 2 = 3a$

$2a^2 - 3a - 2 = 0$ $\quad (2a+1)(a-2) = 0$

$\Downarrow \quad \Downarrow$

$a = -1/2 \quad a = 2$

$\underbrace{e^x = -1/2} \quad e^x = 2 \quad x = \ln 2$

no solution

e^x can't be negative

Ex Show that the inverse of $y=\tanh x$ is $\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

$$y=\tanh^{-1}x \Rightarrow \tanh y = x$$

$$\frac{e^{y}-e^{-y}}{e^{y}+e^{-y}}=x \Rightarrow \frac{e^{2y}-1}{e^{2y}+1}=x$$

$$e^{2y}-1=xe^{2y}+x \qquad e^{2y}(1-x)=x+1$$

$$e^{2y}=\frac{x+1}{1-x} \qquad 2y=\ln\left(\frac{x+1}{x-1}\right) \qquad y=\frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$$

Ex Show that $\cosh^2 x-\sinh^2 x=1$.

$$\cosh^2 x-\sinh^2 x=\left(\frac{e^{x}+e^{-x}}{2}\right)^2-\left(\frac{e^{x}-e^{-x}}{2}\right)^2$$

$$=\frac{1}{4}\left(e^{2x}+2+e^{-2x}\right)-\frac{1}{4}\left(e^{2x}-2+e^{-2x}\right)$$

$$=\frac{1}{4}\left(\cancel{e^{2x}}+2+\cancel{e^{-2x}}-\cancel{e^{2x}}+2-\cancel{e^{-2x}}\right)$$

$$=1.$$

Ex

$$\sinh 2x=\frac{e^{2x}-e^{-2x}}{2}=\frac{(e^{x}+e^{-x})(e^{x}-e^{-x})}{2}$$

$$=\left(\underbrace{\frac{e^{x}+e^{-x}}{2}}_{\cosh x}\cdot\underbrace{\frac{e^{x}-e^{-x}}{2}}_{\sinh x}\right)\cdot 2$$

$$=2\sinh x\cdot\cosh x$$